Rook polynomials and forbidden positions

Combinatorics

Fall 2026

An assignment problem may prohibit some person–position pairings. A rook board keeps those prohibited cells visible while we count only assignments that use allowed cells. The coefficients of a rook polynomial count legal partial assignments; the coefficient for all rows counts legal complete assignments.

After studying these notes, you will be able to do the following.

- Model allowed assignments by white squares of a board and nonattacking rook placements.
- Compute rook numbers and the rook polynomial of a small finite board.
- Use a disjoint-board product and a use-or-omit recurrence to compute a rook polynomial.
- Use the appropriate rook number to count legal complete assignments.
- Count injective assignments when there are more positions than objects.

6.1 Special cases of rook numbers

Definition 6.1 (Rook numbers). Let B be a finite collection of allowed squares in a rectangular grid. For an integer $k \geq 0$, the **kth rook number** $r_k(B)$ is the number of ways to place k nonattacking rooks on squares of B. The placement with no rooks is allowed, so $r_0(B) = 1$.

A board with u rows and v columns has no nonattacking placement of more than $\min\{u, v\}$ rooks. The full boards give a first set of rook numbers that can be counted without examining individual placements.

Example 6.2 (Four rooks on a full 4-by-4 board). How many ways are there to place four nonattacking rooks on a full 4-by-4 board? Explain why these placements are in bijection with the permutations of $\{1, 2, 3, 4\}$.

Solution: A four-rook placement uses four distinct rows and four distinct columns. Since the board has exactly four rows and four columns, it has one rook in every row and one rook in every column. Choose the column for the rook in row 1, then row 2, then row 3, and finally row 4. There are respectively 4, 3, 2, and 1 choices, so

$$r_4(B) = 4 \cdot 3 \cdot 2 \cdot 1 = 4!.$$

The word "equivalently" describes the same collection of placements using functions rather than board squares. Let $\mathcal{R}$ be the set of four-rook placements, and let S_4 be the set of bijective functions from $\{1, 2, 3, 4\}$ to itself. For a placement $P \in \mathcal{R}$, define

$$\pi_P : \{1, 2, 3, 4\} \longrightarrow \{1, 2, 3, 4\}, \qquad \pi_P(i) = \text{the column of the rook in row } i.$$

There is exactly one rook in row i, so $\pi_P(i)$ is defined. Distinct rows have rooks in distinct columns, so different inputs have different outputs; because all four columns are used, π_P is bijective. Conversely, for $\pi \in S_4$, place a rook at $(i, \pi(i))$ for each i. The function assigns one column to each row, and its bijectivity makes those columns distinct, so the rooks are nonattacking. These two instructions undo one another. The left panel of Figure 6.1, for example, records the permutation $\pi = (2, 4, 1, 3)$.

Example 6.3 (Derangements on a 4-by-4 board). Remove the four squares on the main diagonal of a 4-by-4 board. How many ways are there to place four nonattacking rooks on the remaining squares?
Solution: Under the bijection in Example 6.2, the diagonal square (i, i) records the equation $\pi(i) = i$. Thus a placement avoiding every diagonal square corresponds to a permutation with no fixed points: a derangement.
Let A_i be the set of permutations with $\pi(i) = i$. If k specified positions are fixed, the other $4-k$ positions may be permuted in $(4-k)!$ ways. Inclusion–exclusion gives the desired rook number as

$$r_4(B) = 4! - C(4,1)3! + C(4,2)2! - C(4,3)1! + C(4,4)0! = 24 - 24 + 12 - 4 + 1 = 9.$$

Therefore the board without its main diagonal has $!4 = 9$ four-rook placements. The right panel of Figure 6.1 shows one of them.

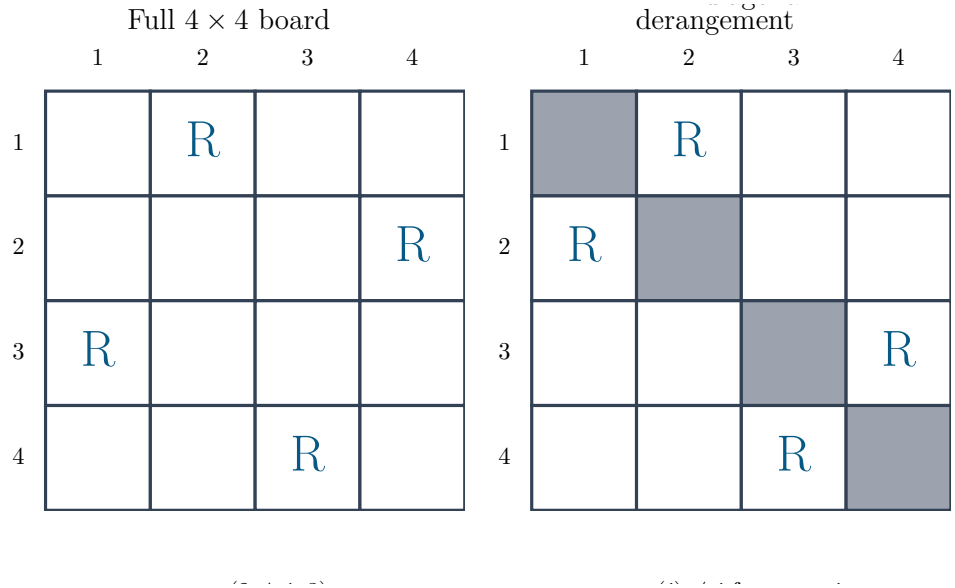


Figure 6.1: A full four-by-four board and the board obtained by removing its main diagonal. A full placement represents a permutation; a full placement avoiding the grey diagonal squares represents a derangement.

Figure description: Two four-by-four boards. The first is full, with one rook in each row and column, and is labeled with the corresponding permutation. The second has grey diagonal squares and four rooks off the diagonal, representing a derangement.

Example 6.4 (Three rooks on a full 5-by-5 board). How many ways are there to place three nonattacking rooks on a full 5-by-5 board?
Solution: A placement has three occupied rows and three occupied columns. First choose its occupied rows, in $C(5,3)$ ways. Next choose its occupied columns, also in $C(5,3)$ ways. These choices leave a 3-by-3 intersection grid, as shown in the first two panels of Figure 6.2.
Each selected row must be paired with a different selected column. Order the three selected rows from top to bottom. The rook in the first selected row has three selected columns available, the next has two, and the final rook has one. Thus the selected intersection grid has 3! nonattacking placements, as the third panel shows. The product rule gives

$$r_3(B) = C(5,3)C(5,3)3! = 10 \cdot 10 \cdot 6 = 600.$$

Every placement determines its occupied rows, its occupied columns, and the matching between them. Conversely, those three choices place the rooks. Hence each three-rook placement is counted exactly once.

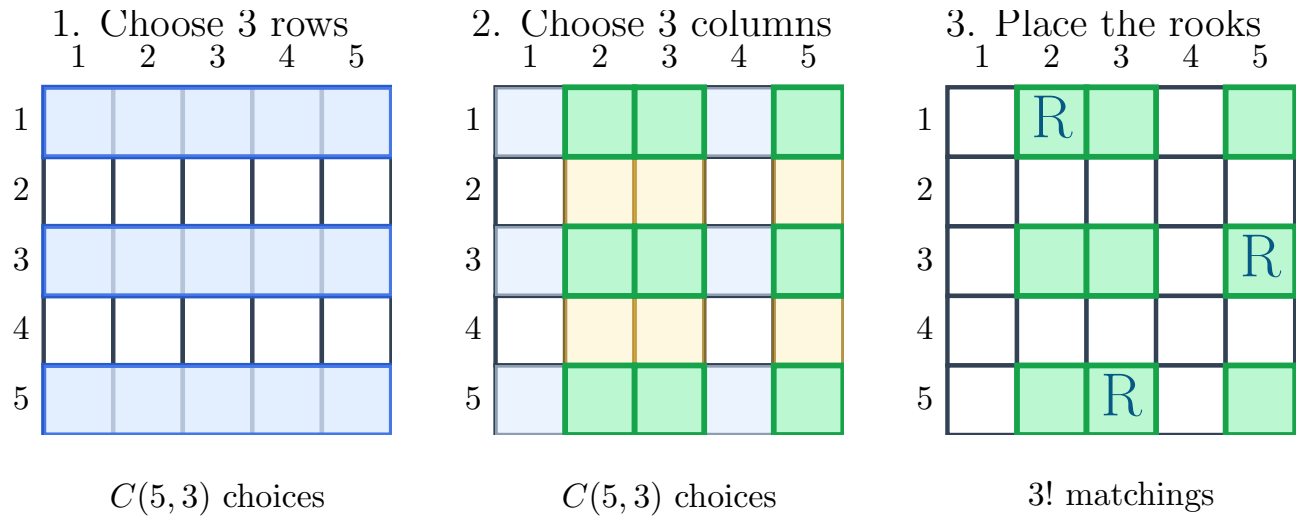


Figure 6.2: Counting three nonattacking rooks on a full five-by-five board. Select three rows, select three columns, and then choose a bijection between the selected rows and columns.

Figure description: Three five-by-five board diagrams. The first highlights rows 1, 3, and 5 in blue. The second marks columns 2, 3, and 5 in yellow; the nine cells at selected-row and selected-column intersections are green. The third shows three rooks in different green intersection cells.

6.2 General boards for restricted assignments

Suppose that three students must be assigned, one each, to three presentation slots. Student i receives slot $\pi(i)$, where π is a permutation of $\{1,2,3\}$. The four pairings in the following display are unavailable:

$$(1,1), \qquad (1,3), \qquad (2,2), \qquad (3,1).$$

For example, $(1,3)$ says that student 1 cannot receive slot 3.

Rows of the grid represent students, and columns represent slots. Grey cells are forbidden. The **allowed board** B consists of the white cells:

$$B = \{(1,2),(2,1),(2,3),(3,2),(3,3)\}.$$

Rooks are placed only on squares of B, never on grey cells. A rook at (i,j) records the legal assignment $\pi(i) = j$. Two rooks are **nonattacking** when they lie in different rows and different columns; then their two assignments can occur together.

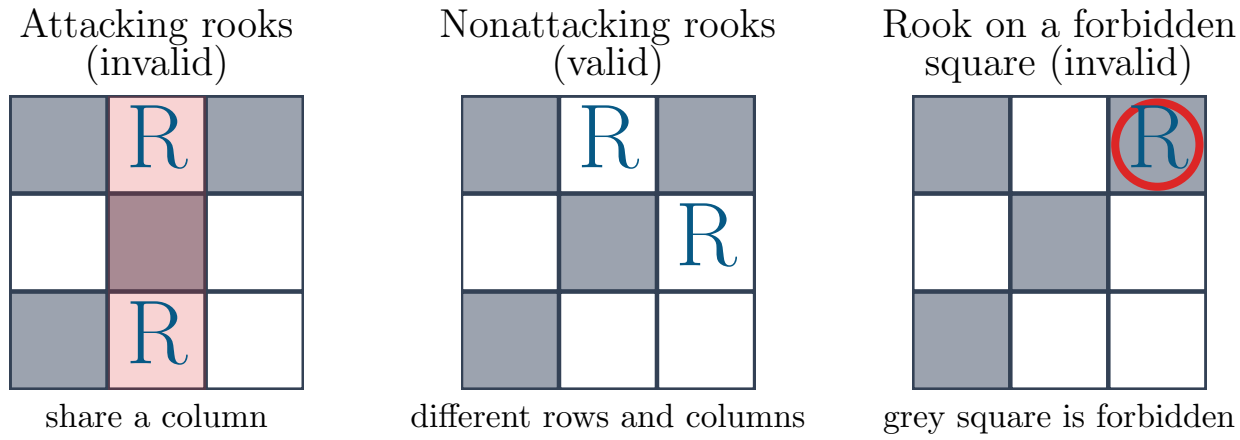


Figure 6.3: The same forbidden-position board in all three panels. Two rooks in one column attack and give an invalid placement. Two rooks in different rows and columns give a valid placement. A rook on a grey forbidden square is invalid.

Figure description: Three copies of the same three-by-three grid. In every copy, cells (1,1), (1,3), (2,2), and (3,1) are grey. The first panel has rooks at (1,2) and (3,2), sharing a column. The second has rooks at (1,2) and (2,3), in different rows and columns. The third has one rook at grey cell (1,3), circled in red.

Rook numbers are computed from the allowed board B, not from the whole assignment grid. Thus every rook counted by $r_k(B)$ lies on a white square. A placement of k nonattacking rooks records k legal assignments made at once. For example, the middle panel of Figure 6.3 records the assignments $\pi(1) = 2$ and $\pi(2) = 3$.

6.3 One board, counted directly

Let B be the allowed board in Figure 6.3. We compute every rook number directly. Every square named in the following list belongs to B.

The count is easier to check when we keep the board, the legal pairs, and the unique three-rook placement in view at the same time. The figure records those objects before the list below explains why each coefficient has its value.

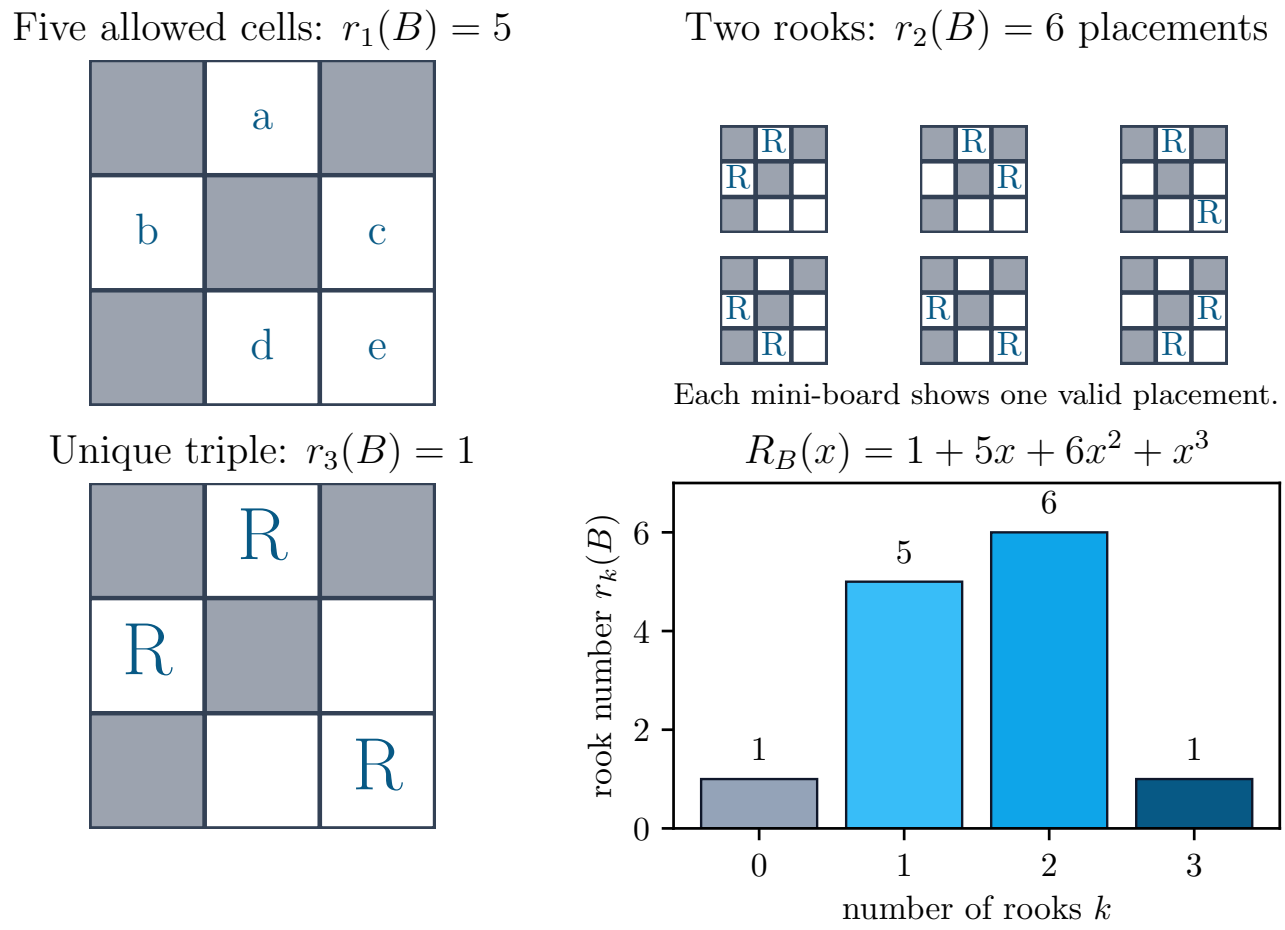


Figure 6.4: A direct count of the rook numbers for B. The five allowed cells give $r_1(B) = 5$; six unordered pairs use different rows and columns; exactly one three-rook placement is possible; and these values become the coefficients of $R_B(x)$.

Figure description: Four panels tracing the direct count. The first labels the five allowed cells of board B. The second shows six scaled-down copies of the board, each with one valid two-rook placement. The third shows the unique three-rook placement. The fourth displays the values 1, 5, 6, and 1 for zero through three rooks and the polynomial 1 plus 5x plus 6x squared plus x cubed.

- There is one zero-rook placement, so $r_0(B) = 1$.
- There are five allowed squares, so $r_1(B) = 5$.
- For two rooks, list the unordered pairs of allowed squares that use different rows and columns:

  $$\{(1,2),(2,1)\}, \quad \{(1,2),(2,3)\}, \quad \{(1,2),(3,3)\}, \quad \{(2,1),(3,2)\}, \quad \{(2,1),(3,3)\}, \quad \{(2,3),(3,2)\}.$$

  Thus $r_2(B) = 6$. A pair such as $\{(2,1),(2,3)\}$ is not counted because its two rooks would share row 2.
- A three-rook placement must use every row and every column. The only possibility is $\{(1,2),(2,1),(3,3)\}$, so $r_3(B) = 1$.

No four-rook placement is possible on a board with only three rows. The list of rook numbers is therefore

$$1, \quad 5, \quad 6, \quad 1, \quad 0, \quad 0, \ldots .$$

6.4 Rook polynomials collect all rook numbers

Counting placements one number at a time repeats the same work whenever we ask about three, four, or five rooks on a larger board. A single polynomial stores all of those counts, with the exponent recording the number of rooks.

Definition 6.5 (Rook polynomial). The **rook polynomial** of a finite allowed board B is

$$R_B(x) = \sum_{k \geq 0} r_k(B) x^k.$$

The coefficient of x^k is the number of legal nonattacking placements of k rooks on B.

For the board B above, direct enumeration gives

$$R_B(x) = 1 + 5x + 6x^2 + x^3.$$

The coefficient of x^2 records the six two-assignment legal placements, and the coefficient of x^3 records the one complete legal assignment.

6.5 Computing rook polynomials without listing every placement

Direct enumeration is reliable for a small board, but it becomes tedious when there are many possible placements. The product rule and the use-or-omit recurrence reorganize the count.

6.5.1 Boards on disjoint rows and columns

Suppose B_1 and B_2 use disjoint sets of rows and disjoint sets of columns. Write $B_1 \sqcup B_2$ for the board formed by taking both sets of squares. A rook placement on this board splits uniquely into a placement on B_1 and a placement on B_2; rooks in one component cannot attack rooks in the other component.

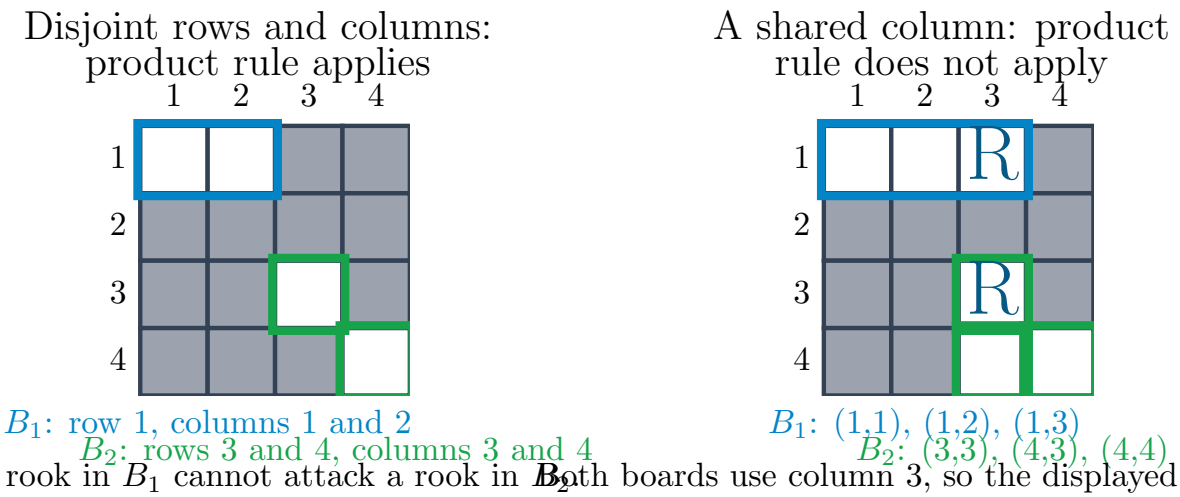


Figure 6.5: The product rule requires both row sets and column sets to be disjoint. At left, a rook placement on B_1 can be combined with any rook placement on B_2. At right, the two subboards share column 3, so placing one rook in each can produce an attacking pair.

Figure description: Two four-by-four board diagrams. In the left grid, blue subboard B1 has white allowed squares at (1,1) and (1,2). Green subboard B2 has white allowed squares at (3,3) and (4,4). The subboards have no row or column in common. In the right grid, blue B1 has white allowed squares at (1,1), (1,2), and (1,3). Green B2 has white allowed squares at (3,3), (4,3), and (4,4). Both subboards use column 3, and rooks at (1,3) and (3,3) attack.

Theorem 6.6 (Product rule for disjoint rook boards). If B_1 and B_2 occupy disjoint row sets and disjoint column sets, then

$$R_{B_1 \sqcup B_2}(x) = R_{B_1}(x) R_{B_2}(x).$$

The right panel of Figure 6.5 shows why both conditions are necessary. Disjoint squares alone are not enough: if two subboards share a row or a column, a rook placement in one may attack a rook placement in the other.

Proof. To form a placement of k rooks on $B_1 \sqcup B_2$, choose an integer j between 0 and k, place j rooks on B_1, and place the remaining $k-j$ rooks on B_2. The decomposition is unique because every square belongs to one component. Therefore

$$r_k(B_1 \sqcup B_2) = \sum_{j=0}^{k} r_j(B_1) r_{k-j}(B_2).$$

The coefficient of x^k in $R_{B_1}(x)R_{B_2}(x)$ is the same sum. Since the two polynomials have the same coefficient for every k, they are equal. □

Example 6.7 (Two separate groups of allowed squares). Suppose B_1 consists of one allowed square. Suppose B_2 consists of two allowed squares in one common row, and all rows and columns of B_2 are disjoint from those of B_1. Then

$$R_{B_1}(x) = 1 + x, \qquad R_{B_2}(x) = 1 + 2x,$$

because the two squares in B_2 attack each other. The combined board has

$$(1 + x)(1 + 2x) = 1 + 3x + 2x^2.$$

There are three one-rook placements and two ways to choose one rook from each component. There is no three-rook placement, as the zero x^3 coefficient also says.

6.5.2 Use or omit an allowed square

Choose an allowed square s of B. Every legal rook placement belongs to exactly one of two classes: it either uses s or leaves s unused. Both classes consist entirely of placements on white allowed squares.

Let $B - s$ denote the allowed board obtained by omitting s. Let $B \mid s$ denote the allowed board remaining after the legal assignment represented by s has been fixed: remove the row and column of s, since no remaining rook may use either one.

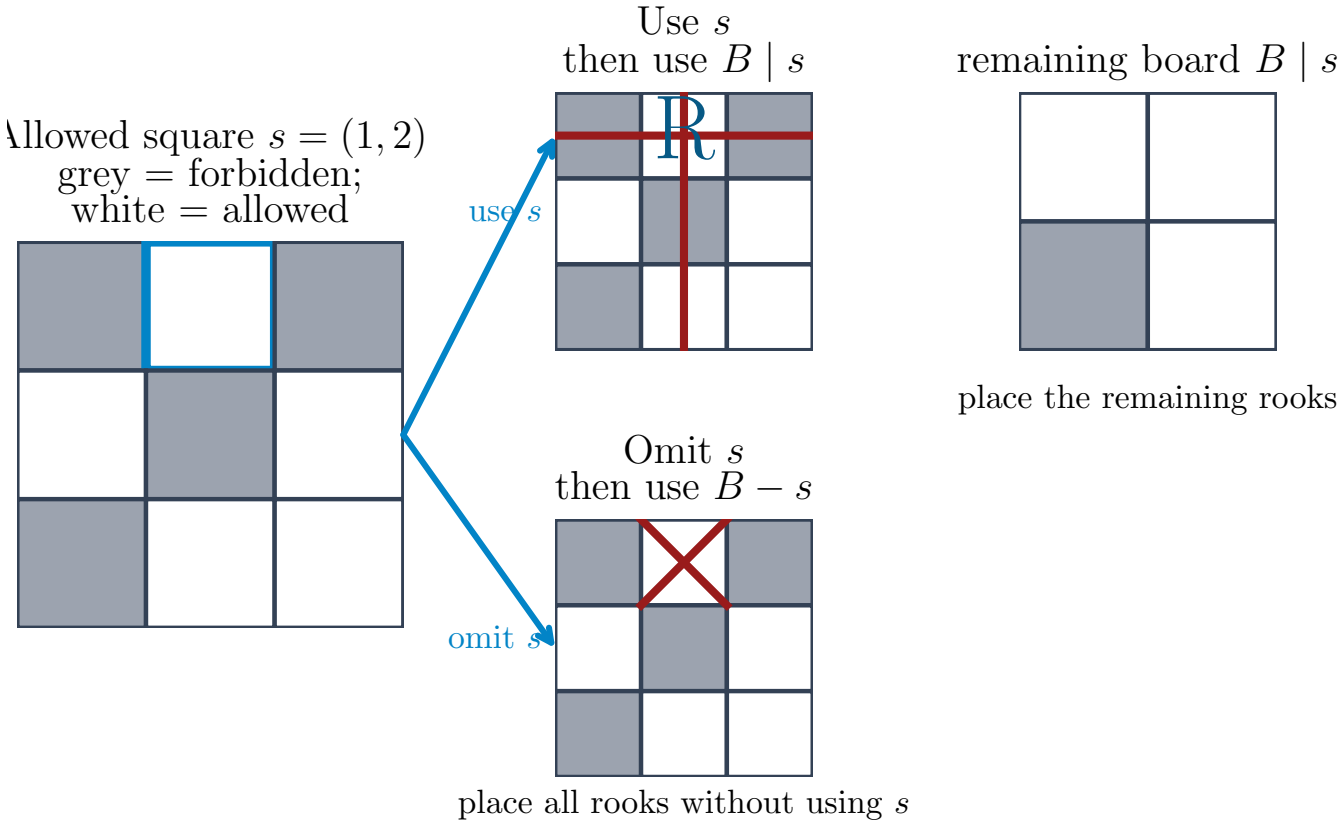


Figure 6.6: The two classes of legal rook placements at the white allowed square $s = (1, 2)$. Using s fixes one legal assignment and leaves the smaller allowed board $B \mid s$. Omitting s leaves the allowed board $B - s$.

Figure description: A branching diagram beginning with the three-by-three board whose grey cells are forbidden and whose white cells are allowed. The white square (1,2) is outlined in blue. The upper branch places a rook there, crosses out its row and column, and shows the remaining two-by-two allowed board. The lower branch crosses out square (1,2) only and shows the original allowed board with that square unavailable.

Theorem 6.8 (Use-or-omit recurrence). For an allowed square s of B,

$$R_B(x) = R_{B-s}(x) + xR_{B|s}(x).$$

Proof. Placements that leave s unused are precisely the placements on $B - s$. For a placement that uses s, remove the rook at s. The remaining rooks form a placement on $B \mid s$, and this operation is reversible by restoring the rook at s. Restoring that rook increases the number of rooks by one, so the second class

contributes $xR_{B|s}(x)$. □

For the board in Figure 6.3, choose the white allowed square $s = (1, 2)$. If that square is left unused, the remaining allowed board has four squares. Its two-rook placements are $\{(2, 1), (3, 2)\}$, $\{(2, 1), (3, 3)\}$, and $\{(2, 3), (3, 2)\}$, so

$$R_{B-s}(x) = 1 + 4x + 3x^2.$$

If a rook is placed at s, the remaining legal assignments may use rows 2 and 3 and columns 1 and 3. The remaining allowed squares are $(2, 1)$, $(2, 3)$, and $(3, 3)$, so

$$R_{B|s}(x) = 1 + 3x + x^2.$$

The use-or-omit recurrence gives

$$\begin{aligned} R_B(x) &= (1 + 4x + 3x^2) + x(1 + 3x + x^2) \\ &= 1 + 5x + 6x^2 + x^3, \end{aligned}$$

which agrees with the direct calculation.

6.6 Legal complete assignments

For an n-by-n allowed board, a placement of n nonattacking rooks uses every row and every column. It is therefore exactly a legal permutation.

Theorem 6.9 (Legal-assignment formula). If B is an allowed board in an n-by-n assignment grid, then the number of legal permutations is $r_n(B) = [x^n]R_B(x)$.

For the board in Figure 6.3, $r_3(B) = 1$. The sole legal permutation is

$$\pi(1) = 2, \qquad \pi(2) = 1, \qquad \pi(3) = 3.$$

6.6.1 More positions than objects

The same reasoning works when n distinct objects are assigned injectively to m distinct positions, where $m \geq n$. A placement of n nonattacking rooks on the allowed board assigns every object to a distinct permitted position.

Theorem 6.10 (Legal injective assignments). If an allowed board B has n rows and m columns with $m \geq n$, then the number of legal injective assignments of the n rows to distinct columns is $r_n(B) = [x^n]R_B(x)$.

Example 6.11 (Three students and four slots). Keep the four forbidden pairings of the grid, but add a fourth presentation slot that is available to every student. The allowed board has three additional squares, one in each row. To find $r_3(B)$, choose the slot for student 1. If student 1 receives slot 2, the other students can be assigned in four ways. If student 1 receives slot 4, they can be assigned in three ways. Thus

$$r_3(B) = 4 + 3 = 7$$

legal injective assignments exist.

6.7 What the polynomial has recorded

The coefficient $[x^k]R_B(x) = r_k(B)$ records the number of legal injective assignments of k rows to k distinct columns. In particular, $r_n(B)$ counts the legal assignments that use every row. The rook polynomial is a finite first example of a generating function: one algebraic object stores a whole sequence of related counts.

Ordinary generating functions extend this bookkeeping idea from the finite sequence $r_0(B), r_1(B), ...$ to arbitrary sequences indexed by size.